MIL-R-25506C
30 August 1982
SUPERSEDING
MIL-R-25506B
24 April 1975

MILITARY SPECIFICATION

RESIN SOLUTION, SILICONE, LOW-PRESSURE LAMINATING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for low-pressure laminating silicone resin used in the fabrication of glass fabric-based plastic laminates that conform to MIL-P-25518 (see 6.1 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-D-705	-	Drum, Shipping and Storage, Steel, 16 and 30 Gallon Capacity.
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55 Gallon (208 Liters).
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).

MILITARY

MIL-P-25518	-	Plastic Materials, Silicone Resin, Glass Fiber Base, Low Pressure Laminated.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9330

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 974 - Neutralization Number by Color-Indicator Titration.
ASTM D 1084 - Viscosity of Adhesives.
ASTM D 1209 - Color of Clear Liquids (Platinum-Cobalt Scale).
ASTM D 1259 - Nonvolatile Content of Resin Solutions.
ASTM D 1298 - Density, Specific Gravity, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The materials used in the manufacture of the resin shall react to form a product which complies with the requirements of this specification. The resin shall be of the silicone, thermosetting, low-pressure laminating type. The component materials of the resin system shall be such that the cured system shall not be corrosive to metals.

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3.2 Form. The resin shall be be dissolved in a suitable solvent and shall be suitable for impregnating fabric. The resin shall be free of foreign matter.

3.3 Code number. The manufacturer shall designate each resin composition by a code number. The code number may include the trade name if desired. The base resin shall be included as part of the code number. Any changes in the components or manufacturing procedures shall necessitate use of a different code number.

3.4 Physical and reactive properties. Physical and reactive properties for the resin furnished under this specification shall be as specified in table I.

3.5 Shelf life. The minimum shelf life of the unactivated resin, when stored at a temperature of 32.2°C (90°F) maximum in the supplier's unopened unit containers, shall be 90 days following the supplier's date of shipment. During this 90 day period, the resin shall be capable of forming laminates conforming to the requirements of the applicable laminate specification.

3.6 Workmanship. The material specified herein shall be processed as required in accordance with the best practice for a high quality resin.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as quality conformance inspections (see 4.3).

4.3 Quality conformance inspection.

4.3.1 Lot formation. Unless otherwise specified herein, a lot shall consist of all the resin of one code number or designation offered for inspection at one time.

4.3.1.1 Sampling.

4.3.1.1.1 Physical property inspection. One gallon of liquid resin or 10 pounds of dry resin shall be selected at random from each lot; also an equivalent quantity of any curing agent which is to be furnished with the resin under the contract (see 6.2.1c). The sample selected shall be examined for all the requirements in table II. Failure of the sample to pass any requirement therein shall be cause to reject the lot represented by the sample.

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4.3.1.1.2 Packaging inspection. The lot size for this inspection shall be the total number of shipping containers. The sample unit shall be one container fully prepared for delivery, just prior to closure. The MIL-STD-105 inspection level shall be S-2 and the AQL shall be 4.0 percent defective. Inspection shall be in accordance with 4.6.

4.4 Test conditions.

4.4.1 Standard conditions. Specimens shall be conditioned and tested at an air temperature of 23° ± 1.1°C (73.4° ± 2°F) and relative humidity of 50 ± 4 percent. The conditioning period, prior to test, shall be 96 hours.

4.5 Test methods.

4.5.1 Specific gravity. The specific gravity of the unactivated and of the cured unfilled resin shall be determined in accordance with ASTM D 1298 or by weighing a specific volume of the resin in grams and dividing by the volume in cubic centimeters.

4.5.2 Viscosity. The viscosity of the liquid unactivated resin shall be determined in accordance with ASTM D 1084.

4.5.3 Nonvolatile content. Nonvolatile content of the resin shall be determined in accordance with ASTM D 1259, method A, using a 1.5 gram (to nearest 0.0001 gram) sample heated 3 hours at 135° ± 1°C (275° ± 2°F). Nonvolatile content shall be determined as follows:

$$\text{Percent nonvolatile content} = \frac{\text{final weight}}{\text{initial weight}} \times 100$$

4.5.4 Gel time. Gel time shall be determined by the manufacturer at 160° ± 1°C (320° ± 2°F).

4.6 Packaging. The sample selected in 4.3.1.1.2 shall be inspected as specified in table III.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or industrial, as specified (see 6.2.1e).

5.1.1 Level A. The resin solution shall be packaged, as specified (see 6.2.1f), in 1-gallon or smaller cans conforming to type V, class 2, round, of PPP-C-96; 5-gallon pails conforming to type II, class 3 of PPP-P-704; 16-gallon drums conforming to type III of PPP-D-705; or 55-gallon drums conforming to type III or type IV of PPP-D-729. Cans in accordance with PPP-C-96 shall be rust-resistant coated on the exterior in accordance with plan B of that specification and the side seam shall be striped with a suitable corrosion-resistant coating.

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5.1.2 Industrial. Packaging shall be such as to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity for immediate use. Providing this requirement is met, packaging in accordance with the supplier's commercial practice shall be acceptable.

5.2. Packing. Packing shall be level A, B or industrial, as specified (see 6.2.1e).

5.2.1 Level A. Resin solution packaged in containers conforming to PPP-C-96 shall be packed in accordance with the overseas requirements of the appendix thereto. Resins packaged in containers conforming to PPP-P-704, PPP-D-705 or PPP-D-729 require no overpacking. When specified (see 6.2.1g), 5-gallon pails and 16-gallon drums shall be palletized in accordance with MIL-STD-147.

5.2.2 Level B. Resin solution packaged in containers conforming to PPP-C-96 shall be packed in accordance with the domestic requirements of the appendix thereto. Resins packaged in containers conforming to PPP-P-704, PPP-D-705, or PPP-D-729 require no overpacking. When specified (see 6.2.1g), 5-gallon pails and 16-gallon drums shall be palletized in accordance with MIL-STD-147.

5.2.3 Industrial. Low pressure laminating resin solution, packaged as specified in 5.1.2, shall be packed to afford protection against corrosion, deterioration and damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking for shipment. In addition to any special marking required by the contract or order (see 6.2.1h) or herein, interior packages, exterior shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129. The nomenclature shall include:

RESIN, SILICONE, LOW-PRESSURE LAMINATING
MIL-R-25506C
Code number (see 3.3)
Date of shipment
Storage conditions
Handling instructions

6. NOTES

6.1 Intended use. Low-pressure silicone resin solution covered by this specification is intended for use in fabricating MIL-P-25518 laminates for general structural parts, aircraft radio and radar antenna housing, and other applications. Laminate requirements are found in MIL-P-25518.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Quantity (weight) (see 6.2.2).

c. Curing agent to be furnished, when applicable (see 6.3).

d. Electrical requirements, when applicable (see table I).

e. Levels of preservation and packing required (see 5.1 and 5.2).

f. Size and type of containers required (see 5.1 and 5.2).

g. Whether palletizing is required (see 5.2.1 and 5.2.2).

h. Special marking, if required (see 5.3).

6.2.2 Purchase unit. The unit of purchase for the resin is the pound.

6.3 Resin systems. The silicone resin described in this specification is used in resin systems comprising the resin itself dissolved in a solvent and a curing agent or agents (hardener, accelerator, etc.), which may constitute a large percentage, by weight, of the cured silicone product. The resins furnished under this specification may be furnished as pre-mixed systems in a partially reacted condition in a single container or the curing agents may be packaged separately to be mixed with the resin at time of use. In the latter case, the packaged curing agents may be acquired with the resin or may be acquired separately. It should be understood that the optimum properties of a given brand of resin may not be developed in the cured product if the curing agents and procedures are not in accordance with the manufacturer's recommendations.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MR
Navy - AS
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Review activities:
Army - AR, AV, MI
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Navy - AS

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TABLE I. Physical and Reactive Properties.

Property	Requirement	Test paragraph
Unactivated base resin:		
Specific gravity	1.08 ± 1	4.5.1
Viscosity at 25°C (77°F), CPS	20 - 50	4.5.2
Color, max	250	ASTM D 1209
Nonvolatile content, percent min	59	4.5.3
Acid number, neutral number, max	0.02	ASTM D 974
Activated resin:		
Gel time at 160°C (320°F), minutes	10 ± 1	4.5.4
Cured, unfilled resin:		
Specific gravity	1.08 ± 1	4.5.1
Dielectric constant	1/	
Loss tangent	1/	

1/ When electrical properties are required (see 6.2.1d), the manufacturer shall furnish nominal values, adequate for design calculations for dielectric constant and loss tangent.

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TABLE II. Physical property inspection. 1/

Property	Paragraph			
	Requirement	Test method	No. of test specimens	Average results to nearest
Specific gravity	3.4	4.5.1	3	0.01 unit
Viscosity	3.4	4.5.2	3	1 unit
Color	3.4	ASTM D 1209	3	2/
Nonvolatile content	3.4	4.5.3	3	1 percent
Acid number	3.4	ASTM D 974	3	2/
Gel time	3.4	4.5.4	3	± 30 seconds

1/ When electrical properties are specified (see table I), test values shall be averaged and reported as specified by the manufacturer.

2/ Results shall be reported as specified in the ASTM method.

TABLE III. Examination of packaging.

Examine	Defect
Preservation	Not level specified in contract requirements. Packaging material not as specified.
Packing	Not level specified in contract requirements. Arrangement not as specified. Any nonconforming packing component, component missing, damaged or otherwise defective affecting serviceability. Closure not as specified.
Count	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

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